Secure Life General Hospital

Problem Statement

The management of a **Secure Life General Hospital** has observed a change in patient admissions, outstanding medical bills, and variations in insurance coverage across its departments. The hospital wants to use patient admission, billing and insurance data to understand its operational performance, monitor payments and improve management decision-making. However, the hospital currently lacks a consolidated report that provides a clear picture of its operational and financial performance.

The hospital management has provided patients dataset containing:

- ***Patient ID****: A unique identification number assigned to each patient visit or record.*
- ***Patient Name****: The name of the patient who received medical care.*
- ***Age****: The patient's age in years at the time of the visit.*
- ***Gender****: The patient's gender, as recorded in the dataset.*
- ***Admission Date****: The date the patient was admitted to the hospital.*
- ***Discharge Date****: The date the patient was discharged from the hospital.*
- ***Department****: The hospital department where the patient received treatment, such as Cardiology, Emergency or Orthopedics.*
- ***Diagnosis****: The medical condition identified or recorded for the patient.*
- ***Total Bill Amount****: The total amount charged by the hospital for the patient's treatment and services.*
- ***Insurance Coverage (%):*** *The percentage of the total bill covered by the patient's health insurance.*
- ***Payment Status****: Indicates whether the patient's bill is Paid, Pending or Partially Paid.*

You have been engaged as a Data Analyst to analyse the dataset. Your task is to transform the raw data into meaningful insights and build an interactive dashboard that will help hospital management monitor performance and make informed decisions. ***Also, your analysis shouldn't be limited to the objectives below. Other insights that can enhance decision making should be added.***

Project Objectives

Your analysis should help management answer the following questions:

1. How many patient visits has the hospital recorded?
2. Which departments receive the highest number of patients?
3. Which departments generate the highest total bills?
4. What is the average length of stay for patients across different departments?
5. What proportion of patient bills has been paid, remains pending, or has been partially paid?
6. How much has the hospital billed, and how much remains outstanding?
7. How does insurance coverage vary across departments and diagnoses?
8. Which age groups account for the highest number of patient visits?
9. How do admissions and billing amounts change over time?
10. Which areas may require management attention based on the analysis?

Note*: Each row represents one patient visit, not necessarily one unique patient. A patient may visit the hospital multiple times and therefore appear in more than one row..*